Leibniz Institute of Ecological Urban and Regional Development

ILUS - International Land Use Symposium 2025

Spatial dynamics in an urban society

6 & 7 November 2025 in Dresden, Germany

Programme

Conference Agenda

Overview and details of the sessions of this conference. Please select a date or location to show only sessions at that day or location. Please select a single session for detailed view (with abstracts and downloads if available).

Titles **Authors** samaniego

Sessions including 'samaniego'

US4A: Urban Structure and Policy: Scaling

Time: **Thursday, 06/Nov/2025: 5:45pm - 6:45pm** *Location:* **Frauenkirche I**
Session Chair: **Horacio Samaniego**

Presentations including 'samaniego'

Socioeconomic Segregation in Voicecall Networks: Analyzing the Impact of City Size in Chile and Brazil

Horacio Samaniego[1], Joaquin Villagra[2], Yerka Freire[3], Alexandre Evsukof[4]
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Since the late 1990s, research on activity spaces has shifted the focus of segregation studies be yond residential areas, incorporating both temporal and spatial dimensions [1]. These expanded measures of segregation complement traditional residential segregation indices by capturing real-world social interactions. This shift enables a transition from place-based to people-based measures of segregation, offering a more dynamic understanding of social divides. Early work by Hägerstrand introduced a methodological framework to explicitly define the geography of social interactions, inspiring a steady stream of research across the social and natural sciences [2]. His contributions elevated this framework into a distinct sub-discipline known as Time Geography. Although initially overlooked due to the significant challenges of data management and analysis, Hägerstrand's approach has gained substantial traction in recent years. This resurgence is largely driven by advancements in technology, which have simplified the aggregation and analysis of individual trajectories, contextual factors, and infrastructural constraints across urban spaces [3]. For example, Alessandretti et al. [4] demonstrate, using mobile phone data, that despite the exploratory nature of human behavior, individuals tend to confine their activity spaces to a limited number of locations. This finding aligns with Dunbar's assertion of a cognitive limit—approximately 100 stable social relationships that an individual can maintain [5]. In this study, we explore how segregation, measured through voice call interactions, varies across cities of different sizes and socioeconomic status (SES) using mobile phone data in Chile and Brazil. By merging SES information with a Call Detail Records (CDR) dataset from a major mobile phone provider, we assess segregation patterns in voice call interactions across a gradient of city sizes in Chile. We analyze 12.5 and 1,258 million anonymized voice calls in Chile and Brazil respectively (representing social events) between reliable users to construct voice call networks for cities of different sizes. These networks are tagged with the SES of callers, call duration, and urban distance between callers, allowing us to characterize the nature of directed interactions based on SES. The SES of each user is estimated by determining their most probable residential location. Segregation is evaluated using a modified exposure index (E) applied across SES quin tiles, providing insights into the dynamics of social interactions within and between socioeconomic groups. More specifically, we evaluate E between SES quintiles as following: Eαβ = N/ NαNβ ∑T (ntα ntβ / nt) [6]. Here, α and β are SES, N total population. T is defined as the social space constructed from the egonet of diameter equal to three. These egonets were built for each user with known SES in the network. While city networks are generally sparselikely due to the dataset representing only a fraction of the market shareseveral network indices help describe the social structure of voice call interactions across cities and over time. For instance, call reciprocity remains large and consistent across cities. Assortativity by SES is positive in all networks, indicating that users tend to connect with others of similar socioeconomic status.

Session Details:

SN3: Urban Dynamics in Global South and Global North
Time: 07/Nov/2025: 11:30am-1:00pm · Location: Frauenkirche II

Further information

Home

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ILUS 2025 at a glance

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2025
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